

Prescribing the ice–regolith contact angle

Wall free energy, its weak form, and why no surface energy survives into the residual coefficient

lunar_regolith_DSM

2026-09-16

Summary. The substrate in this two-phase model is the domain wall, not a phase field, so it must act through a boundary condition. Cahn’s wall free energy $f_w(\phi) = \gamma_{as} + (\gamma_{is} - \gamma_{as})h(\phi)$ joins the bulk functional as an ordinary surface term — both are energies, both in $\text{J}/\text{m}^2 \times \text{m}$, and they add directly. Its variation supplies a natural condition on $\partial_n \phi$ that fixes the contact angle. The resulting residual term is $-3M \cos \theta h'(\phi)/6$, with no ϵ and no surface energy in the coefficient. Both cancellations are derived below, and neither requires a unit conversion.

1. The free energy

Let ϕ be the ice fraction, $\phi = 1$ ice and $\phi = 0$ vapour, and let $\Gamma \subseteq \partial\Omega$ be the part of the boundary representing regolith. Write the whole free energy physically, in energy units throughout:

$$\mathcal{F}[\phi] = \underbrace{\int_{\Omega} \left[3\gamma_{ia}\epsilon |\nabla\phi|^2 + \frac{3\gamma_{ia}}{\epsilon} \phi^2(1-\phi)^2 \right] d\Omega}_{\mathcal{F}_{\text{bulk}}} + \underbrace{\oint_{\Gamma} f_w(\phi) d\Gamma}_{\mathcal{F}_{\text{wall}}}, \quad (1)$$

$$f_w(\phi) = \gamma_{as}(1 - h(\phi)) + \gamma_{is}h(\phi) = \gamma_{as} + (\gamma_{is} - \gamma_{as})h(\phi), \quad h(\phi) = \phi^2(3 - 2\phi). \quad (2)$$

Both terms of (1) are energies — in 2D, J per unit depth — so they add with no conversion between them. This is worth stating because it is easy to reach for one: the form of $\mathcal{F}_{\text{bulk}}$ implemented in the code (§3) is non-dimensionalised and is *not* an energy, and mixing that form with a physical f_w does require a conversion. Writing both physically, as here, avoids the question entirely.

The bulk coefficients in (1) are fixed by requiring that the ice–vapour interface carry surface energy γ_{ia} . For the equilibrium profile

$$\phi(x) = \frac{1}{2} \left(1 + \tanh \frac{x}{2\epsilon} \right), \quad \phi' = \frac{\phi(1-\phi)}{\epsilon}, \quad (3)$$

the two bulk terms are equal (equipartition, from the first integral of the Euler–Lagrange equation), and

$$\int_{-\infty}^{\infty} \left[3\gamma_{ia}\epsilon \phi'^2 + \frac{3\gamma_{ia}}{\epsilon} \phi^2(1-\phi)^2 \right] dx = 6\gamma_{ia}\epsilon \int \phi'^2 dx = 6\gamma_{ia}\epsilon \cdot \frac{1}{6\epsilon} = \gamma_{ia}. \quad (4)$$

(Verified numerically: ratio 1.00000000.) So (1) is calibrated, and ϵ sets the interface width without changing its energy.

Two properties of h matter. $h(0) = 0$, $h(1) = 1$ give $f_w(0) = \gamma_{as}$ (wall against vapour) and $f_w(1) = \gamma_{is}$ (wall against ice). And $h'(0) = h'(1) = 0$ makes the wall term inert in both bulk phases: it acts only where the ice–vapour interface meets the wall, and cannot shift the bulk equilibria. A linear interpolant would violate the second property and bias ϕ all along Γ .

Note that (2) contains no γ_{ia} — only the two substrate energies. γ_{ia} appears in $\mathcal{F}_{\text{bulk}}$, and the ratio of the two is what produces $\cos \theta$ in §2.1.

2. First variation and the weak form

Vary (1) with an arbitrary test function η :

$$\delta\mathcal{F}[\eta] = \int_{\Omega} \left[6\gamma_{ia}\epsilon \nabla\phi \cdot \nabla\eta + \frac{6\gamma_{ia}}{\epsilon} f_1(\phi) \eta \right] d\Omega + \oint_{\Gamma} f'_w(\phi) \eta d\Gamma, \quad (5)$$

where

$$f_1(\phi) = \frac{1}{2} \frac{d}{d\phi} [\phi^2(1-\phi)^2] = \phi(1-\phi)(1-2\phi) \quad (6)$$

is the half-normalised double-well derivative the code uses — the normalisation that makes (3) have width exactly ϵ .

Equation (5) is the weak form: the volume integral is never integrated by parts in the Galerkin discretisation, and the boundary integral enters alongside it directly. That is the whole mechanism — the wall term is a surface integral in the residual, nothing more.

2.1 Strong form and the natural boundary condition

To see what boundary condition (5) encodes, integrate the gradient term by parts (for this purpose only):

$$\int_{\Omega} 6\gamma_{ia}\epsilon \nabla\phi \cdot \nabla\eta d\Omega = - \int_{\Omega} 6\gamma_{ia}\epsilon \nabla^2\phi \eta d\Omega + \oint_{\partial\Omega} 6\gamma_{ia}\epsilon (\partial_n\phi) \eta dS. \quad (7)$$

Substituting into (5) and collecting,

$$\delta\mathcal{F}[\eta] = \int_{\Omega} \left[-6\gamma_{ia}\epsilon \nabla^2\phi + \frac{6\gamma_{ia}}{\epsilon} f_1 \right] \eta d\Omega + \oint_{\partial\Omega} \left[6\gamma_{ia}\epsilon \partial_n\phi + f'_w(\phi) \right] \eta dS. \quad (8)$$

η is arbitrary in Ω and independently arbitrary on $\partial\Omega$, so both brackets vanish. The second is the natural boundary condition:

$$6\gamma_{ia}\epsilon \partial_n\phi + f'_w(\phi) = 0. \quad (9)$$

Where no wall term is declared, $f'_w \equiv 0$ and (9) reduces to $\partial_n\phi = 0$ — the zero-flux condition the solver had everywhere before this feature, which is a 90° contact angle imposed by omission.

With $f'_w = (\gamma_{is} - \gamma_{as})h'$ and $h'(\phi) = 6\phi(1-\phi)$,

$$\partial_n\phi = \frac{\gamma_{as} - \gamma_{is}}{6\gamma_{ia}\epsilon} h'(\phi) = \underbrace{\frac{\gamma_{as} - \gamma_{is}}{\gamma_{ia}}}_{\cos\theta} \frac{\phi(1-\phi)}{\epsilon}. \quad (10)$$

Young's equation was not assumed. The wall energy contributed only $\gamma_{as} - \gamma_{is}$; the γ_{ia} came from the bulk coefficient in (1). Their ratio is Young's $\cos\theta$, and it appears because the two terms of the same functional were each calibrated independently — the bulk to γ_{ia} , the wall to γ_{as} and γ_{is} .

By (3) the right-hand side of (10) is $\cos\theta |\nabla\phi|$, so writing $\nabla\phi = |\nabla\phi| \mathbf{m}$ with $\mathbf{m}$ the interface normal into the ice,

$$\boxed{\mathbf{m} \cdot \mathbf{n} = \cos\theta} \quad (11)$$

with $\mathbf{n}$ the outward wall normal: the contact angle measured through the ice. Limits: $\cos\theta = +1$ gives $\mathbf{m} = \mathbf{n}$, ice sheeted flat onto the wall, $\theta = 0$; $\cos\theta = -1$ gives $\theta = 180^\circ$, point contact.

3. The residual, and the form the code uses

The evolution is the gradient flow $\partial_t \phi = -M_{\text{phys}} \delta \mathcal{F} / \delta \phi$, so with $\eta = N_a$ the residual is

$$R_a = \int_{\Omega} N_a \phi_t d\Omega + M_{\text{phys}} \left\{ \int_{\Omega} \left[6\gamma_{ia} \epsilon \nabla N_a \cdot \nabla \phi + \frac{6\gamma_{ia}}{\epsilon} f_1 N_a \right] d\Omega + \oint_{\Gamma} f'_w(\phi) N_a d\Gamma \right\}. \quad (12)$$

The code stores a single mobility `mob_sub` = M , calibrated through $d_{0,\text{sub}0} \rightarrow \lambda_{\text{sub}} \rightarrow \tau_{\text{sub}}$ to deliver a requested kinetic coefficient β . Matching (12) against the implemented interior form

$$R_a = \int N_a \phi_t + 3M\epsilon \int \nabla N_a \cdot \nabla \phi + \frac{3M}{\epsilon} \int f_1 N_a - (\text{source}) \quad (13)$$

gives $M_{\text{phys}} \cdot 6\gamma_{ia} = 3M$, i.e.

$$\boxed{M_{\text{phys}} = \frac{M}{2\gamma_{ia}}} \quad (14)$$

which reproduces *both* bulk coefficients simultaneously — a consistency check, since one number is fixed by two equations.

The boundary term of (12) then follows immediately:

$$R_a^{\Gamma} = M_{\text{phys}} (\gamma_{is} - \gamma_{as}) h'(\phi) N_a = \frac{M}{2\gamma_{ia}} (\gamma_{is} - \gamma_{as}) h'(\phi) N_a = \boxed{-3M \cos \theta \frac{h'(\phi)}{6} N_a} \quad (15)$$

and, since the wall energy has no ϕ_t dependence, a Jacobian with no shift term,

$$J_{ab}^{\Gamma} = -3M \cos \theta \frac{h''(\phi)}{6} N_a N_b, \quad h''(\phi) = 6(1 - 2\phi). \quad (16)$$

Equation (15) is `assembly.c` verbatim: `Rb[a][0] = -rwb * 3.0 * mob_b * costhet * (dh/6.0) * N0b[a]`.

3.1 Why no surface energy survives

γ_{ia} cancels in (15) between f'_w and M_{phys} , and it does so for a reason that is not bookkeeping: **the mobility is calibrated to a kinetic coefficient, not to an energy**. M is whatever makes the interface move at the requested β ; by (14) that forces $M_{\text{phys}} \propto 1/\gamma_{ia}$, so no energy scale can survive into any residual coefficient. Had γ_{ia} remained, that would have been the error.

The ϵ cancels for the same structural reason: f'_w carries none, and M_{phys} carries none, so (15) cannot.

3.2 An equivalent route, and where $\epsilon/6$ comes from

Some presentations non-dimensionalise the bulk term first, writing

$$\tilde{\mathcal{F}} = \int_{\Omega} \left[\frac{\epsilon^2}{2} |\nabla \phi|^2 + W(\phi) \right] d\Omega, \quad W = \frac{1}{2} \phi^2 (1 - \phi)^2, \quad \mathcal{F}_{\text{bulk}} = \frac{6\gamma_{ia}}{\epsilon} \tilde{\mathcal{F}}, \quad (17)$$

because $\tilde{\mathcal{F}}$'s interfacial excess is $\epsilon/6$ — a *length* — rather than an energy. The flow is then $\partial_t \phi = -(3M/\epsilon) \delta \tilde{\mathcal{F}} / \delta \phi$, which is where the familiar $3M/\epsilon$ prefactor comes from.

Taking that route, the wall term must be converted into the same non-dimensional currency, $\tilde{f}_w = (\epsilon/6\gamma_{ia}) f_w$, and the $\epsilon/6$ reappears explicitly. It gives the same answer, because $(3M/\epsilon) \cdot (\epsilon/6\gamma_{ia}) = M/(2\gamma_{ia}) = M_{\text{phys}}$.

The route in §3 is preferable precisely because it never mixes the two currencies. A conversion factor is only needed if one term is written physically and the other is not.

4. Clamping, and why it is required

$h'(\phi) = 6\phi(1 - \phi)$ changes sign for $\phi < 0$ and $\phi > 1$. So does (15), and since $R = N\phi_t + \dots = 0$, a sign-flipped wall term drives ϕ *further* out of range: a runaway.

Nothing in the bulk opposes it. The interior form evaluates f_1 and the localiser at the clamped $\phi_c \in [0, 1]$, so for $\phi < 0$ both are identically zero — there is no restoring force outside the range by construction. An unclamped wall term is then the only term acting there, and it acts outward.

This is not hypothetical. At $\epsilon/R = 1/100$ a -2×10^{-6} excursion at the wall grew to -0.05 within a few hundred steps, crossed `-phase_lo`, and the residual’s domain guard then returned a zero residual, which SNES read as $\|F\| = 0$ and reported as converged. The solution froze for the remaining 23,370 steps while the clock ran on to t_{final} and the run reported success. The implementation therefore clamps ϕ to $[0, 1]$ before evaluating h' , with the Jacobian clamping identically and carrying the chain-rule factor $d\phi_c/d\phi$, so that it remains an exact Jacobian of the clamped residual.

5. Using it

<code>-wall_faces</code>	<code>"y0,y1"</code>	which domain faces are regolith
<code>-gamma_is</code>		ice–regolith surface energy [J/m ²]
<code>-gamma_as</code>		air–regolith surface energy [J/m ²]
<code>-gamma_ia</code>		ice–air; default $(\Sigma_i + \Sigma_a)/2$

The solver forms $\cos \theta = (\gamma_{as} - \gamma_{is})/\gamma_{ia}$, refuses the run if $|\gamma_{as} - \gamma_{is}| > \gamma_{ia}$ (no angle exists; one phase wets completely), and prints both the energies and the derived θ .

On the default. γ_{ia} defaults to $(\Sigma_i + \Sigma_a)/2$, not to Σ_i . In the convention these projects inherit, $\Sigma_i = \gamma_{ia} + \gamma_{is} - \gamma_{as}$ and $\Sigma_a = \gamma_{ia} + \gamma_{as} - \gamma_{is}$, so $\Sigma_i + \Sigma_a = 2\gamma_{ia}$ and the ice–air energy is their mean — which is also what the solver’s own double-well coefficient $C = (\Sigma_i + \Sigma_a)/2 + \Lambda$ says. Σ_i alone equals γ_{ia} only when $\Sigma_a = \Sigma_i$, which `enceladus_DSM` enforces and `lunar_regolith_DSM` does not; with the lunar defaults (0.109, 0.132) taking Σ_i understates γ_{ia} by 10.6%, and that propagates directly into θ . Pass `-gamma_ia` explicitly whenever the substrate matters.

The only admissibility constraint is $|\gamma_{as} - \gamma_{is}| \leq \gamma_{ia}$. The three-phase triple-well constraint $\gamma_{is} > \frac{\gamma_{ia}}{2}(1 - \cos \theta)$ does *not* apply: it came from requiring a positive Σ_s , and there is no sediment phase and no triple well here.

6. Validation

Five prescribed angles from 30° to 150° in a flat channel recover Young’s prediction with a maximum error of 0.127° and an RMS of 0.062° , against a 2° acceptance criterion, with the error falling as ϵ/R is refined. A droplet on a single wall, which has no confinement, agrees with the channel, so the angle is set by the boundary condition rather than by the geometry. Details in `studies/contact_angle/`.